

IoT Smart Irrigation and Plant Monitoring System

Objective

To monitor soil moisture, temperature, and humidity using ESP32 and automatically control irrigation when soil becomes dry.

Features

- Soil moisture monitoring
- Temperature monitoring
- Humidity monitoring
- OLED display output
- Automatic pump control
- Smart agriculture application

Hardware

- ESP32 (38 Pin)
- DHT11 Sensor
- Soil Moisture Sensor
- OLED Display
- Relay Module
- Mini Water Pump

Software

- Arduino IDE
- ESP32 Board Package
- DHT Library
- Adafruit SSD1306 Library